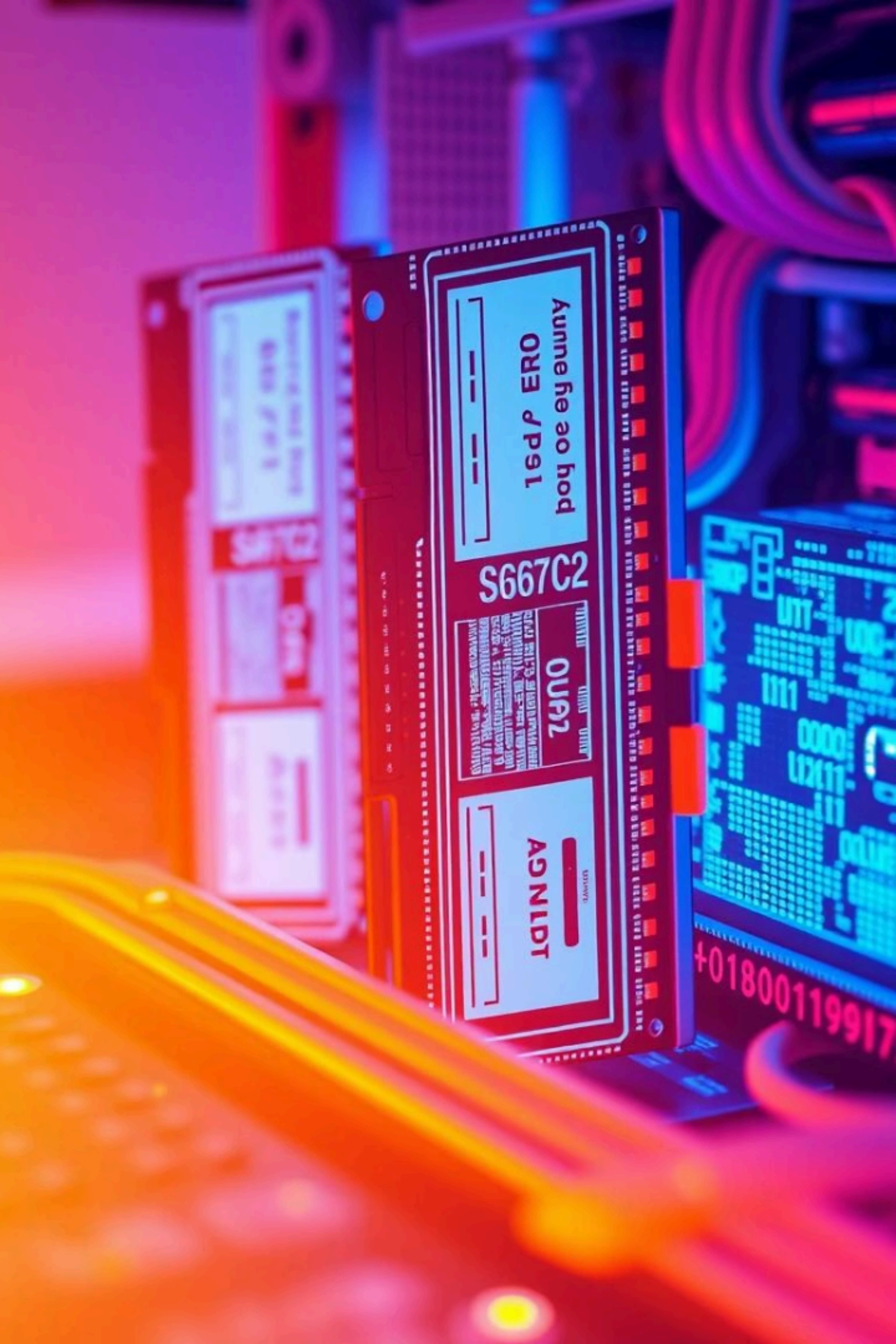




Predictive Page Replacement using Machine Learning

This project report presents an intelligent memory management solution that enhances system performance by leveraging machine learning for page replacement decisions. Unlike traditional static algorithms such as LRU or FIFO, this approach uses predictive analytics to forecast future page references, reducing page faults and improving responsiveness. The system dynamically adapts to varying workloads, making smarter memory management decisions in real-time.

Project Overview and Objectives

Intelligent Page Replacement

Develops a mechanism that predicts upcoming memory access patterns using machine learning models to reduce page faults and improve memory utilisation.

Dynamic Adaptation

Unlike static algorithms, it adapts to changing workloads, enhancing system responsiveness and efficiency.

Performance Improvement

Reduces disk I/O by anticipating required pages, leading to faster application performance and better resource use.



Key Features and Innovations

ML-Based Page Prediction

Learns from past memory access to forecast future page needs.

Live Visualization

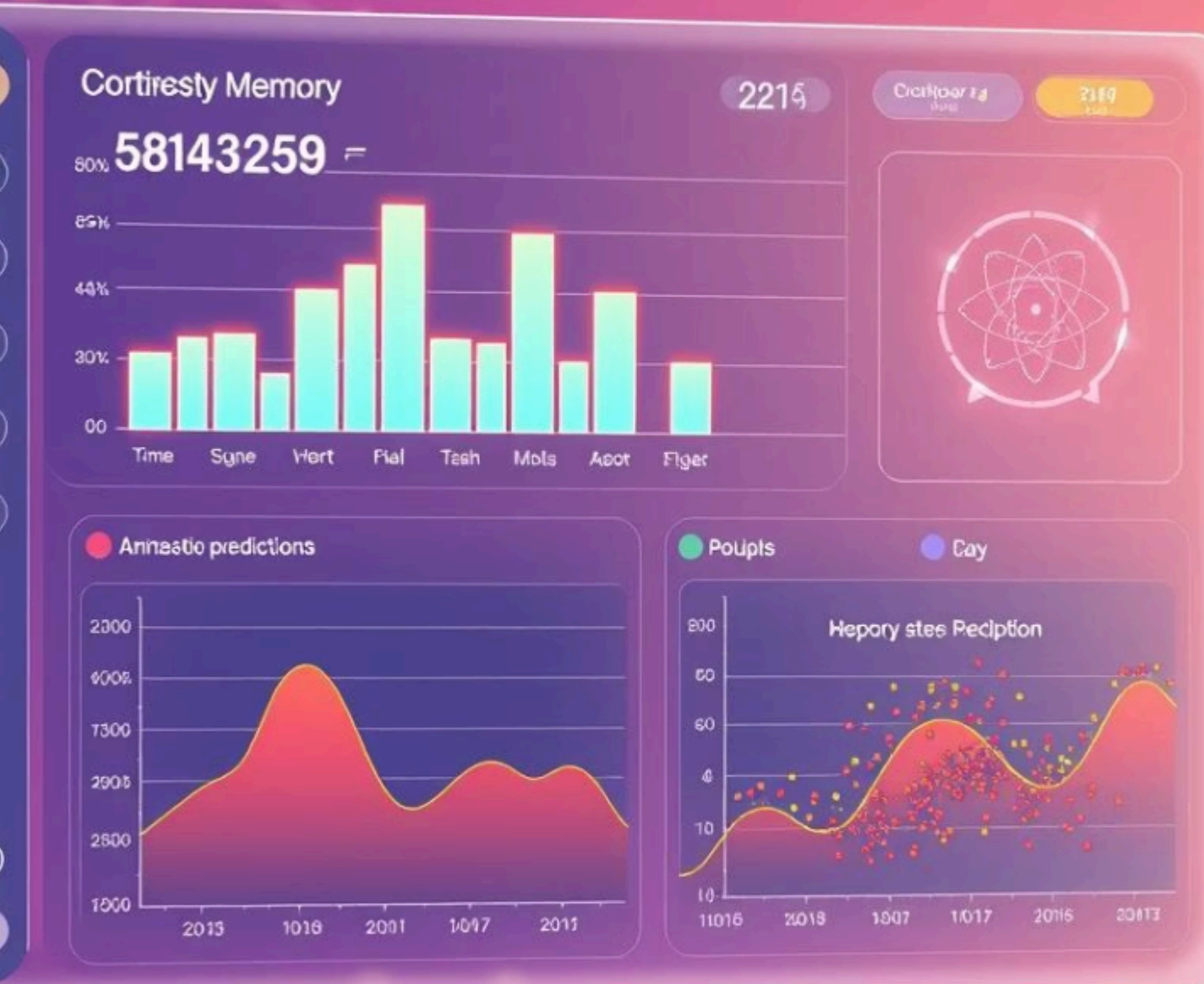
Displays memory usage, prediction results, and comparisons with traditional algorithms.

Custom ML Models

Supports various algorithms like decision trees, SVM, and neural networks for experimentation.

Performance Benchmarking

Shows hit/miss rates and memory efficiency metrics for different strategies.



Use Cases and Target Users

Operating System Developers

Test and integrate intelligent memory management into OS kernels.

Students & Researchers

Study machine learning impact on OS resource management.

System Architects & Engineers

Design efficient embedded and real-time systems with optimized memory.

Cloud Providers & AI Practitioners

Reduce memory overhead and evaluate ML models in system-level use cases.



Module-Wise Breakdown

Module Name	Description	Technology Used
Memory Access Tracker	Captures real-time page access sequences and logs memory usage patterns.	Python (psutil, memory_profiler)
Page Replacement Simulator	Simulates traditional algorithms for benchmarking.	Python (Matplotlib, Tkinter)
ML-Based Prediction Engine	Trains ML models to predict next page access.	Scikit-learn, Pandas, NumPy
Visualization Dashboard	Displays real-time stats and memory insights.	Streamlit / Flask, Plotly

Functionalities and Technology Stack

Functionalities

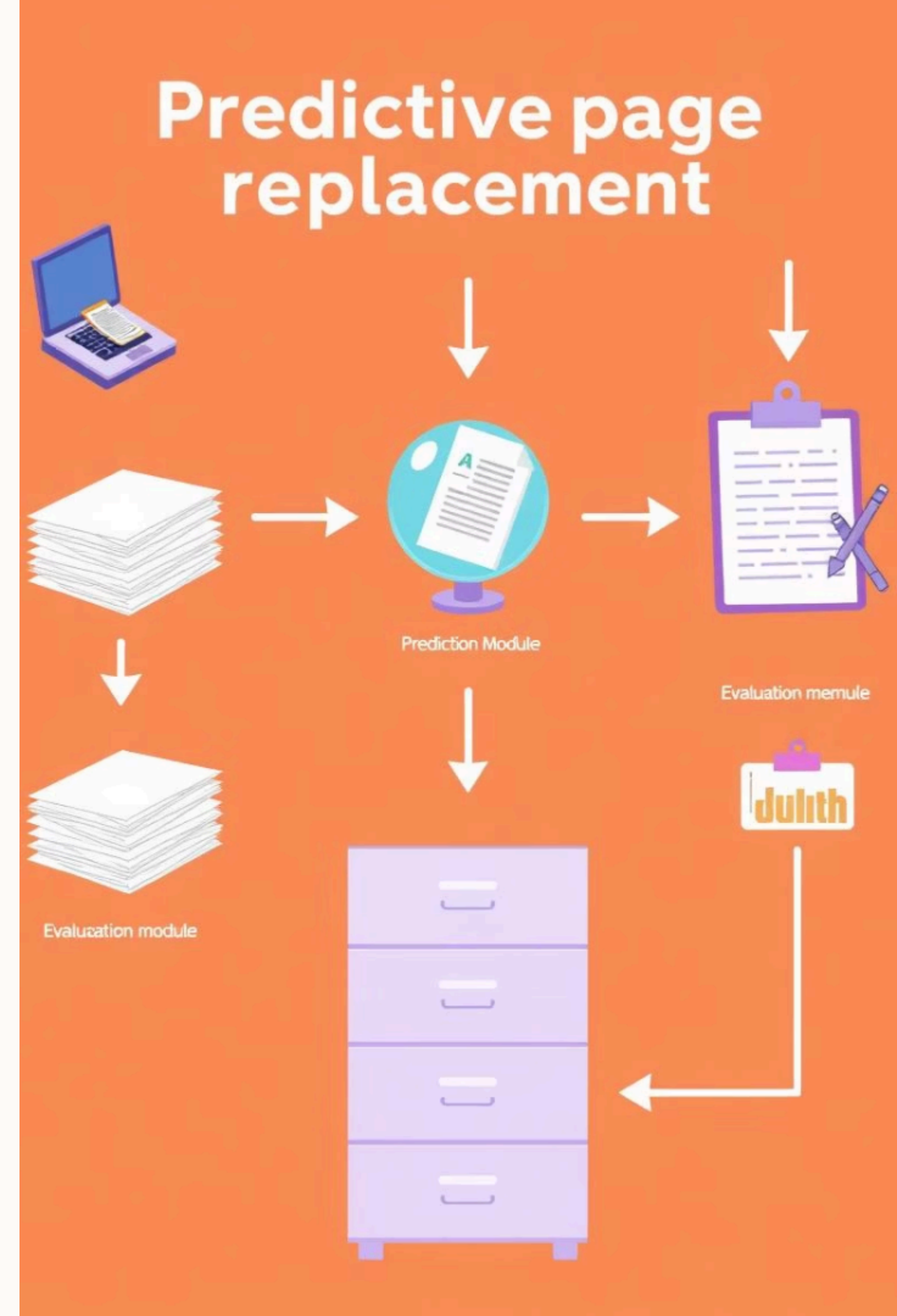
- Real-time memory access logging
- Simulation of traditional algorithms (FIFO, LRU, LFU)
- ML-based page prediction and dynamic replacement
- Accuracy monitoring and performance metrics
- Visualization dashboard and custom model training

Technology Used

- Python for backend and ML implementation
- JavaScript for frontend interactivity
- HTML/CSS for dashboard structure and styling

Flow Diagram and System Architecture

This flow diagram illustrates the interaction between modules: memory access logging, ML prediction engine, and visualization dashboard. It shows how data is collected, processed, and used to make intelligent page replacement decisions, enabling dynamic adaptation and performance improvement.



Conclusion and Future Scope

1

Conclusion

Machine learning enables dynamic page replacement that forecasts future memory access, reducing page faults and improving responsiveness compared to static algorithms.

2

Future Enhancements

- Integration with real OS or virtual machines for practical testing
- Support for deep learning models like LSTM for sequential prediction
- Web-based live simulation and visualization interface
- Extension to cache memory and hybrid paging systems

